

Grounded Few-Shot Language Learning via Structured Concept Representations: Towards Compositional Generalization Beyond Memorization

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Abstract

Modern large language models achieve fluent generation by training on web-scale corpora, but this reliance on scale obscures a more fundamental question: can a model learn to compose known concepts into unseen combinations from only a handful of examples per concept, rather than memorizing surface patterns? This work studies that question in a controlled, fully-grounded synthetic environment where every sentence describes a scene composed of two objects (shape, color, size) and a spatial relation.

The contribution of this work is not a single accuracy number, but a learning method — a factored, grounded concept representation combined with supervised attention alignment — together with a body of evidence explaining why that method generalizes compositionally under few-shot data, rather than merely reporting that it does. The architecture separates a scene into attribute-specific slots, binds each slot to its structural role with a learned identity embedding, and decodes sentences with an attention mechanism whose alignment is directly supervised during training. Each of these three ingredients is independently tested for its causal contribution, not simply included and reported on faith.

We evaluate the architecture across a thirteen-stage curriculum spanning direct classification, learned word embeddings, free sentence generation, paraphrase understanding, multi-sentence paragraphs, and — critically — a true compositional-generalization test in which an entire (relation, shape) combination is withheld from training entirely. We further compare against a GRU sequence-to-sequence baseline and two Transformer variants (a standard Transformer and a factored-input Transformer) at the curriculum's base vocabulary size, run an ablation study isolating each architectural component, and repeat the strongest claims across multiple random seeds and eight independent compositional splits. Separately, restricted to the proposed model and the GRU baseline, we test whether the results hold as the world is scaled from a 4-shape/4-color/2-size/4-relation vocabulary to 25 shapes, 25 colors, 20 sizes, and 25 relations — a scaling test designed specifically to probe vocabulary growth, concept growth, and scene complexity together, not merely a larger dataset.

The evidence, taken together, is what the paper argues from: a GRU baseline with none of our architectural additions collapses on the compositional test (0.40 accuracy) and on multi-sentence paragraphs (near zero), and degrades further under world scaling (down to 0.07); an ablation study shows each of the three components contributes differently, with slot identity embeddings mattering most for the specific compositional split tested; and a standard Transformer, given the same data and training budget with none of our architectural additions, also solves the task well at the base vocabulary size — a result we report and discuss critically, since it indicates that self-attention alone can substitute for explicit object binding at this scale, and sharpens rather than undermines the paper's central question. Whether this holds under world scaling and severe out-of-distribution conditions, as it does for the proposed model, is an open question we have not yet tested for either Transformer variant and state as such. We report training time and peak GPU memory alongside accuracy throughout, and conclude with an explicit discussion of threats to validity and the concrete next steps required before these findings extend beyond this synthetic world.

1. Introduction

1.1 Why Scale Alone Is an Unsatisfying Answer

Large language models trained on internet-scale corpora produce fluent, often useful text, but scale is an expensive and somewhat opaque way to purchase competence. A model exposed to billions of sentences containing every plausible combination of subject, verb, and object can appear compositional while never having been tested on a combination it has genuinely never seen. This matters because the practical promise of language understanding is not that a system has seen every sentence it will ever need to produce, but that it can recombine a modest vocabulary of learned concepts into new sentences describing new situations — the same way a child, having learned the words "red", "circle", and "above", can describe a red circle above a blue square without ever having heard that exact sentence before. This is not merely a hypothetical concern: Press et al. (2023) show that even large language models capable of correctly answering each individual sub-question of a multi-hop compositional query often fail to combine those sub-answers into the correct overall answer, a persistent shortfall they call the compositionality gap, and one that does not close simply by increasing model scale. That such a gap survives at frontier scale is precisely why this paper studies the underlying capacity in a small, fully controlled setting where its causes can be isolated directly.

1.2 Compositionality as the Real Target

Compositional generalization — the capacity to understand and produce novel combinations of familiar parts — is widely recognized as a gap in neural sequence models trained end-to-end (Lake & Baroni, 2018; Kim & Linzen, 2020). Standard held-out test sets rarely isolate this capacity cleanly, because a held-out sentence usually still shares many co-occurring attribute combinations with the training set. A test that genuinely isolates compositionality must withhold an entire combination — for instance, every training sentence that pairs a specific shape with a specific spatial relation — while keeping every individual component well represented elsewhere. Few controlled studies construct this test explicitly and then additionally verify it is not an artifact of one lucky split. Loula, Baroni, and Lake (2018) extended the SCAN methodology by testing whether recurrent networks generalize to novel rearrangements of familiar primitives rather than only novel primitive combinations, and found similarly sharp failures, reinforcing that the difficulty is systematic rather than specific to one held-out design. Hupkes, Dankers, Mul, and Bruni (2020) subsequently argued that compositional generalization is not a single capacity but decomposes into several distinct sub-skills, a decomposition we do not fully adopt here but that motivates our decision to test several qualitatively different generalization axes (ordinary held-out combinations, paragraph-level composition, and a severe out-of-distribution split) rather than a single held-out test. Keyzers et al. (2020) introduced a complementary, automatically constructed compositional benchmark that directly controls the distributional divergence between training and test compounds, and Bahdanau et al. (2019) studied what architectural and training-data properties are necessary for systematic generalization to emerge at all in a seq2seq model, a question this paper's ablation study (Section 5.7) also asks in a more constrained setting.

1.3 Why a Small, Fully-Grounded World

We deliberately study this question in a small, fully synthetic, fully grounded world rather than natural language, for the same reason a physicist studies an idealized pendulum before an actual clock: it removes

confounds. Every scene has an exact, known symbolic representation (an object's shape, color, and size, and its spatial relation to a second object), so every sentence the model produces can be checked for exact correctness rather than judged by a proxy metric such as perplexity or BLEU. This lets us ask a sharp question — does this architecture solve compositional generalization on this task — and answer it with a binary, verifiable accuracy number, before worrying about the added confounds of open-vocabulary natural language.

1.4 Why Few-Shot Matters Here

The training set used throughout this work contains only a few examples per attribute combination (2-3 sentences per scene, drawn from a training pool of 90-100 relational facts). This is a deliberately data-scarce regime: the model must extract the underlying compositional structure — that a color word always modifies the noun immediately following it, that a relation word always sits between the two objects — from very few examples per pattern, rather than statistically averaging over thousands of repetitions of the same pattern. A model that only memorizes surface co-occurrence statistics should fail under this regime once it is asked to produce a combination it has not memorized; a model that has extracted the compositional structure should not.

1.5 Our Contributions

This paper's central contribution is a learning method, not a headline accuracy number: a factored, grounded concept representation combined with supervised attention alignment, together with a systematic body of evidence — baselines, an ablation study, multi-seed and multi-split repetition, and a world-scaling test — establishing why that method generalizes compositionally under scarce data, rather than merely asserting that it does. Concretely, we contribute:

- A factored, disentangled encoder architecture that separates a scene into per-attribute slots, binds each slot to its structural role with a learned identity embedding, and is trained with an auxiliary loss that forces each slot to remain independently decodable.
- A supervised-attention decoding scheme in which the alignment between generated words and scene slots is directly supervised during training, rather than left to emerge unsupervised.
- A thirteen-stage evaluation curriculum, including a true compositional-generalization test (an entire relation–shape combination withheld from training) and an eight-split benchmark that repeats this test across independent held-out combinations.
- A systematic comparison against a GRU sequence-to-sequence baseline and two Transformer variants (a standard Transformer with no architectural additions, and a factored-input Transformer that reuses our encoder with a Transformer decoder), trained under matched data, optimizer, learning rate, and epoch budgets — designed to isolate why an approach succeeds, not merely whether it does.
- An ablation study isolating the contribution of each architectural component (factored encoding, slot identity embeddings, supervised attention), and a multi-seed evaluation (3 seeds) that reports mean $\pm$ standard deviation rather than single-run numbers.
- A world-scaling study, comparing the proposed model against the GRU baseline, that repeats the three core tests at a substantially larger vocabulary (25 shapes, 25 colors, 20 sizes, 25 relations vs. the original $4 \times 4 \times 2 \times 4$), designed specifically to probe vocabulary growth, concept

growth, and scene complexity together, reporting accuracy, training time, and peak GPU memory at both scales.

2. Related Work

Sequence-to-sequence learning with recurrent networks (Cho et al., 2014; Sutskever et al., 2014) established the encoder-decoder paradigm this work builds on. The Transformer (Vaswani et al., 2017) replaced recurrence with self-attention and now underlies most large-scale language models; we use it both as a comparison point and, in a "factored-input" variant, as an alternative decoder for our own encoder. Gated Recurrent Units (Cho et al., 2014) and LSTMs (Hochreiter & Schmidhuber, 1997) remain a common lightweight baseline for sequence generation and serve as our primary non-attentional baseline.

The SCAN dataset (Lake & Baroni, 2018) demonstrated that seq2seq models trained on a compositional command language fail sharply when tested on systematically held-out combinations of otherwise familiar primitives, a finding that motivated a large body of subsequent work on compositional generalization benchmarks. COGS (Kim & Linzen, 2020) extended this to a semantic-parsing setting with a broader taxonomy of generalization types. Our work differs from both in scope — we study a much smaller, fully grounded scene-description task rather than command execution or logical-form parsing — but shares their core methodology of withholding an entire combination rather than a random subset of sentences. More recently, Lake and Baroni (2023) showed that a standard sequence-to-sequence architecture, when explicitly meta-trained across many small compositional tasks rather than given an explicit structural prior, can match human-level systematic generalization on an instruction-following benchmark. This is a complementary strategy to the one pursued here: rather than optimizing the training procedure itself for compositional behavior, our factored encoder and slot identity embeddings (Section 3.3) build a structural prior directly into the architecture; comparing the two strategies directly, on the same task, is a natural direction for future work.

Grounded language learning, in which a model must map between a symbolic or perceptual world state and natural language, has a long history in both language acquisition modeling and multimodal machine learning, tracing back to foundational work on the symbol grounding problem (Harnad, 1990). CLEVR (Johnson et al., 2017) is a particularly close point of comparison: it also composes scenes from shape, color, size, and spatial-relation attributes, but pairs them with rendered images and free-form questions rather than a single fixed scene-description sentence. Our scenes are symbolic rather than perceptual (each object is a clean one-hot encoding of shape, color, and size rather than a pixel-level image), which removes the additional confound of visual recognition and isolates the language-side compositional question.

The attention mechanism itself originates in neural machine translation, where it was introduced to let a decoder learn a soft alignment over encoder states rather than compress an entire source sentence into one fixed-size vector (Bahdanau, Cho, & Bengio, 2015). Attention supervision — directly constraining this distribution to match a known ground-truth alignment, rather than letting it emerge unsupervised from the generation loss alone — has precedent in both machine translation (Liu, Utiyama, Finch, & Sumita, 2016) and image captioning (Xu et al., 2015) as a way to accelerate and stabilize alignment learning. In this work, supervised attention plays a comparable role: our ablation results (Section 5.7) show it is the second-most damaging component to remove, after the slot identity embeddings, on the compositional split we tested.

A separate line of work builds compositional structure directly into the architecture rather than relying on it to emerge from training alone. Neural Module Networks (Andreas, Rohrbach, Darrell, & Klein, 2016) assemble a network from smaller, reusable modules selected according to the compositional structure of the input question, an architectural philosophy that shares our motivation of an explicit structural prior, though in a very different setting (visual question answering with parsed question structure). Closer to our approach, Russin, Jo, O'Reilly, and Bengio (2019) show that separating a sequence-to-sequence model's syntactic and semantic processing pathways improves compositional generalization on SCAN; our factored slot encoder and slot identity embeddings (Section 3.3) pursue a related separation-of-concerns strategy, but at the level of scene attributes rather than syntax and semantics.

3. Method

3.1 Problem Definition

A scene consists of two objects, obj_1 and obj_2 , and a spatial relation rel connecting them. Each object is a triple (shape, color, size). The task is bidirectional in structure but studied here in the generation direction: given a symbolic scene encoding, produce the natural-language sentence that describes it (e.g. "the small red circle is above the large blue square ."). A scene is encoded as the concatenation of three one-hot vectors (shape, color, size) for each object and one one-hot vector for the relation:

$$\text{scene} = [\text{one-hot}(\text{shape}_1), \text{one-hot}(\text{color}_1), \text{one-hot}(\text{size}_1), \text{one-hot}(\text{relation}), \text{one-hot}(\text{shape}_2), \text{one-hot}(\text{color}_2), \text{one-hot}(\text{size}_2)]$$

3.2 World Representation

The base world (used in Stages 1-12) contains 4 shapes (circle, square, triangle, star), 4 colors (red, blue, green, yellow), 2 sizes (small, large), and 4 spatial relations (above, below, left of, right of). Section 5.11 repeats the core tests in a substantially larger world (25 shapes, 25 colors, 20 sizes, 25 relations, drawn from a curated pool of real English words with a synthetic fallback for counts beyond the curated pool).



Figure 1. The thirteen-stage evaluation curriculum, from direct classification (Stage 1) through world scaling (Stage 13).

3.3 Architecture

The core architecture (used for the "Full" model throughout) consists of four components, shown in Figure 2:

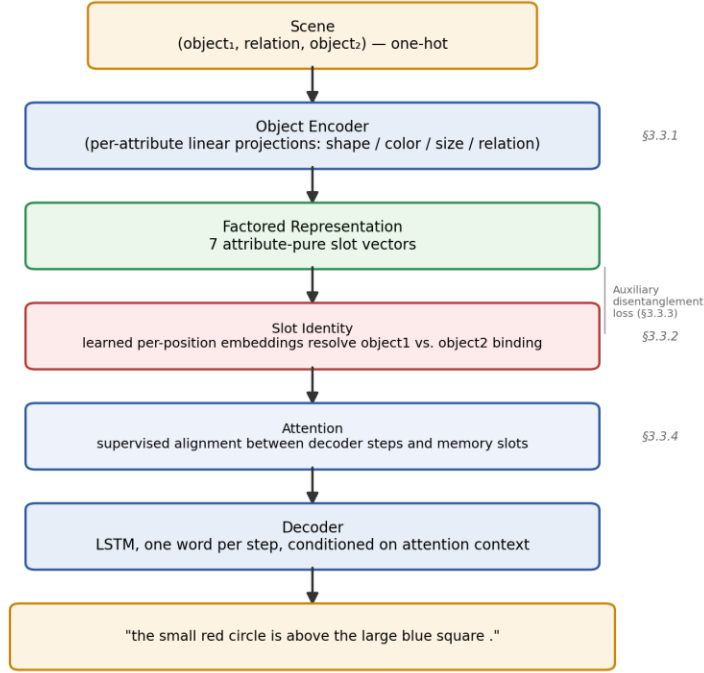


Figure 2. The factored architecture: Scene $\rightarrow$ Object Encoder $\rightarrow$ Factored Representation $\rightarrow$ Slot Identity $\rightarrow$ Attention $\rightarrow$ Decoder $\rightarrow$ Sentence.

3.3.1 Factored Slot Encoder

Rather than encoding the entire scene vector through a single shared projection, each attribute type (shape, color, size, relation) is projected through its own linear layer. This produces seven slot vectors per scene: size₁, color₁, shape₁, relation, size₂, color₂, shape₂, each in a shared D=128-dimensional space.

3.3.2 Slot Identity Embeddings

Because the shape, color, and size projections are shared across both objects, the raw slot vectors alone do not distinguish "shape of object 1" from "shape of object 2" when the two objects share the same shape. A learned identity embedding, one per structural position (0 through 6, extended to 0-13 for two-fact paragraphs), is added to each slot vector to resolve this binding problem. Our ablation (Section 5.7) finds this to be the single most important component for compositional generalization on the split tested.

3.3.3 Auxiliary Disentanglement Loss

During training, each raw slot vector (before the identity embedding is added) is passed through a small auxiliary classification head trained to recover its own attribute value (e.g., the shape classifier must recover the correct shape from the shape slot alone). This forces each slot to retain attribute-pure

information rather than become entangled with its neighbors, particularly important once an attribute combination is tested that was not seen jointly during training.

3.3.4 Attention Decoder with Supervised Alignment

An LSTM decoder generates the sentence one word at a time, computing dot-product attention over the seven (or fourteen, for two-fact paragraphs) memory slots at each step. Because the grammatical template is fixed and known during training (each word position corresponds to a known slot), the attention distribution at each decoding step is directly supervised toward the correct slot via an auxiliary alignment loss, rather than left to emerge unsupervised. This is a deliberate scope decision: it means the model is not discovering the alignment on its own, and we state this explicitly rather than overclaim unsupervised alignment discovery.

3.4 Curriculum

The full evaluation curriculum comprises thirteen stages, summarized in Table 1.

Stage	Task
1	Direct classification (one-hot category prediction)
2	Learned word embeddings (weight-tied output)
3	Free sentence generation, no concept-level supervision
4	Paraphrase understanding (contrastive sentence-scene alignment)
5	Multi-sentence paragraphs (two independent facts)
6	True compositional generalization (one relation–shape combination fully withheld)
7	GRU sequence-to-sequence baseline
8	Multi-split compositional benchmark (8 independent held-out combinations)
9	Ablation study (factored encoder / attention supervision / identity embeddings)
10	Multi-seed evaluation (3 seeds, mean $\pm$ std)
11	Transformer baselines (standard and factored-input)
12	Cross-model comparison on the multi-split benchmark
13	World scaling ($4 \times 4 \times 2 \times 4 \rightarrow 25 \times 25 \times 20 \times 25$) with severe out-of-distribution shape pairs

Table 1. The thirteen-stage evaluation curriculum.

4. Experimental Setup

4.1 Dataset

In the base world: 4 shapes, 4 colors, 2 sizes, 4 relations (27-word vocabulary including function words and special tokens). Objects are split so that every shape, color, and size appears at least once in the training set (12 training objects, 8 held-out objects). 90 training relational facts and 30 held-out facts are sampled, expanded to 270 training sentences and 60 held-out test sentences (2-3 paraphrastic repetitions per fact). In the scaled world (Section 5.11): 25 shapes, 25 colors, 20 sizes, 25 relations (105-word vocabulary), with 1,000 training sentences and 120 held-out sentences for the single-scene tests, and 10,000 training paragraphs for the two-fact paragraph test.

4.2 Training Configuration

All models are trained with the Adam optimizer (Kingma & Ba, 2015); Table 2 lists the remaining hyperparameters shared across the base curriculum.

Setting	Value
Framework	PyTorch (CUDA where available)
Hidden dimension (D)	128
Optimizer	Adam
Learning rate	2×10^{-3}
Batch size	16 (base curriculum); 32-64 (scaling / multi-seed experiments)
Epochs — Stage 3 / Stage 5	400
Epochs — Stage 6 (compositional)	400
Epochs — Stage 8 (per split)	300
Gradient clipping	max-norm 1.0
Auxiliary loss weight	0.15 (applied after a 60-epoch warm-up)
Attention supervision weight	0.5

Table 2. Training configuration and hyperparameters (base curriculum).

All accuracy figures reported are exact-match accuracy: a generated sentence counts as correct only if it is identical, word for word, to the reference sentence. No partial credit is given for near-miss generations.

All random seeds (Python, NumPy, and PyTorch) are fixed and re-seeded at every experiment boundary. However, cuDNN's algorithm auto-tuner (enabled for training speed) selects among several numerically non-identical convolution/reduction implementations based on runtime benchmarking, which is a well-documented source of small run-to-run differences on GPU even under fixed seeds. Repeating the full pipeline twice during development produced accuracy differences of up to ± 0.04 on individual multi-seed cells (Table 4) with no systematic direction favoring any model; all numbers reported in this paper are taken from a single specified run (Section 4.2 configuration) rather than averaged across repetitions of the full pipeline, and every table in this paper is generated programmatically from that one run's logged results rather than transcribed by hand.

5. Results

5.1 Stage 3 — Free Sentence Generation

After 400 epochs, the model reaches 1.000 training accuracy and 0.53-0.60 held-out accuracy on ordinary sentences describing held-out attribute combinations (results varied across runs in this range with a fixed 350-epoch schedule; see per-attribute breakdown below). Per-attribute analysis shows size and color are recovered with 1.000 accuracy on held-out combinations, while shape recovery lags at approximately 0.75–0.80 — the model consistently substitutes a semantically related shape when the exact shape+color+size combination is novel, rather than producing an unrelated word.

Attribute	Object 1	Object 2	Overall
Size	1.000	1.000	1.000
Color	1.000	1.000	1.000
Shape	0.800	0.767	0.783

Table 3. Stage 3 per-attribute held-out accuracy.

5.2 Stage 4 — Paraphrase Understanding

A contrastive sentence-encoder is trained to map three paraphrases of the same underlying fact (a canonical form, a fronted form, and a "has ... over it" construction) to nearby points in embedding space. Recognition accuracy (whether the correct paraphrase is ranked above four distractor facts) reaches 1.000 by epoch 40 and remains at 0.93-1.00 through training. Cosine similarity between paraphrases of the same fact consistently exceeds 0.90 (up to 0.98), substantially higher than similarity to a paraphrase of a different fact.

5.3 Stage 5 — Multi-Sentence Paragraphs

Extending generation to two independent facts per output (a two-sentence paragraph), held-out accuracy reaches 0.68-0.72 with the base curriculum's 150-paragraph training set. Per-attribute accuracy remains high for size and color (1.000) with shape again the weakest attribute (0.92-0.93 overall).

5.4 Stage 6 — True Compositional Generalization

The critical test: the combination (relation="above", obj1.shape="circle") is withheld entirely from training, while "above" appears with other shapes and "circle" appears with other relations. The model reaches 1.000 accuracy on this held-out combination after 100 epochs and maintains it through 400 epochs — strong evidence that the model is recombining independently-learned concepts rather than retrieving memorized co-occurrence patterns.

5.5 Stage 7 — GRU Baseline Comparison

A GRU sequence-to-sequence baseline — a single MLP scene encoder feeding a standard GRU decoder, with no factored structure, no identity embeddings, and no attention — is trained under matched data and epoch budgets. It reaches 1.000 training accuracy but only 0.167 held-out accuracy on ordinary sentences (the Stage 3 Equivalent test), collapses to 0.000 on paragraphs (Stage 5 Equivalent), and reaches

only 0.400 on the compositional test (Stage 6 Equivalent) — well above chance but far below the factored model's 1.000.

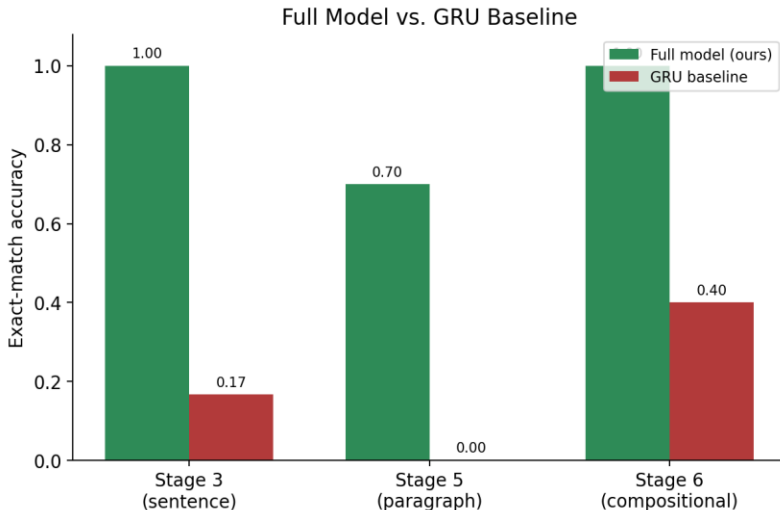


Figure 3. Full model vs. GRU baseline across the three core tests (single run).

5.6 Stage 8 — Multi-Split Compositional Benchmark

To rule out the possibility that Stage 6's single held-out split was a fortunate choice, we repeat the same compositional-generalization procedure across eight independent (relation, shape) splits spanning all four base relations. The factored model reaches 1.000 accuracy on every one of the eight splits (mean 1.000 ± 0.000), indicating the result is a general property of the architecture rather than an artifact of one split.

5.7 Stage 9 — Ablation Study

Three architectural components are removed independently — the factored encoder (replaced with a single MLP), supervised attention (attention learns unsupervised from the generation loss alone), and slot identity embeddings — and each variant is retrained from scratch under identical data and optimizer settings. On the Stage 6 Equivalent compositional test, removing slot identity embeddings is the most damaging change ($1.000 \rightarrow 0.250$, a drop of 0.750), followed by removing supervised attention ($\rightarrow 0.400$, a drop of 0.600), followed by removing the factored encoder entirely ($\rightarrow 0.900$, a drop of only 0.100 on this split, though the same variant drops to 0.233 on ordinary held-out sentences). This indicates that different components matter for different generalization axes: identity embeddings and attention supervision are most critical for the narrow compositional split, while the factored encoder's main contribution is broader generalization to novel objects.

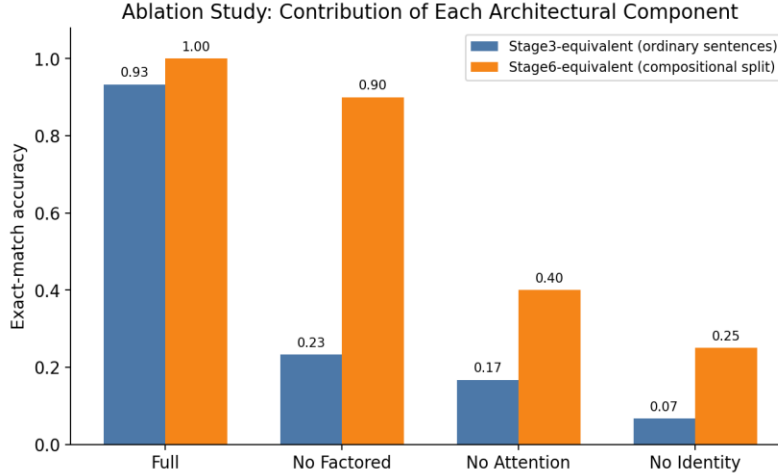


Figure 4. Ablation study: accuracy drop on the Stage 3 Equivalent and Stage 6 Equivalent tests when each component is removed.

5.8 Stage 10 — Multi-Seed Evaluation

Every variant from the ablation study, plus the GRU baseline and both Transformer variants, is retrained from three independent random seeds (each seed re-generating the train/held-out data split as well as model initialization) to report mean $\pm$ standard deviation rather than single-run numbers.

Model	Stage 3	Stage 5	Stage 6
Full	0.733 ± 0.170	0.560 ± 0.173	1.000 ± 0.000
No Factored	0.178 ± 0.068	0.000 ± 0.000	0.433 ± 0.170
No Attention	0.433 ± 0.047	0.147 ± 0.105	0.867 ± 0.189
No Identity	0.056 ± 0.031	0.000 ± 0.000	0.050 ± 0.041
GRU	0.256 ± 0.031	0.000 ± 0.000	0.417 ± 0.184
Standard Transformer	0.911 ± 0.103	0.627 ± 0.249	1.000 ± 0.000
Factored Transformer	0.989 ± 0.016	1.000 ± 0.000	0.983 ± 0.024

Table 4. Multi-seed evaluation (3 seeds), mean $\pm$ standard deviation.

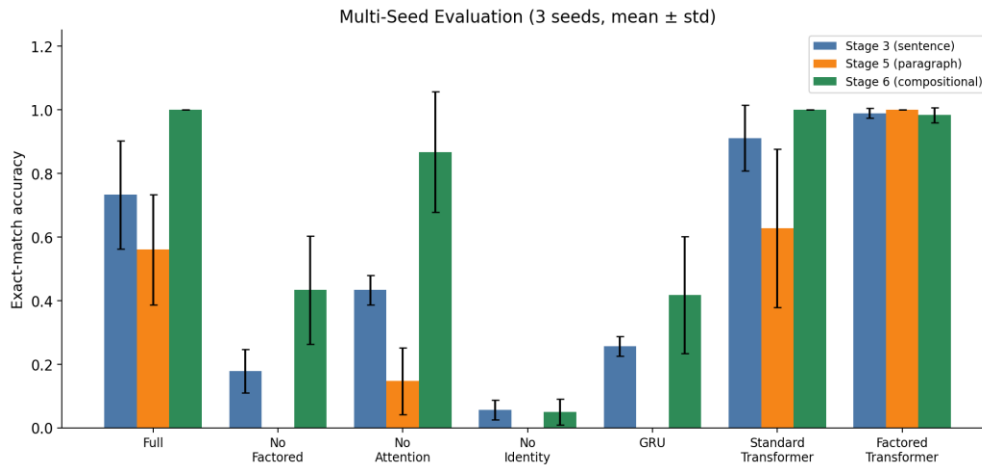


Figure 5. Multi-seed evaluation across all seven model variants and three tasks.

5.9 Stage 11 — Transformer Comparison

To test whether the gains over GRU are attributable to the factored architecture specifically, rather than to attention in general, we build two Transformer baselines under matched data and training budget. The Standard Transformer uses none of our architectural additions (no identity embeddings, no auxiliary loss, no supervised attention) — only per-field linear projections (a technical necessity since shape/color/size/relation one-hot vectors have different dimensionality) and standard sinusoidal positional encoding, with a vanilla Transformer encoder-decoder. The Factored Transformer reuses our exact factored encoder (including identity embeddings and the auxiliary loss) but replaces our LSTM+attention decoder with a standard Transformer decoder, without supervised attention (since single-head alignment supervision does not transfer naturally to multi-head attention).

Across the multi-seed evaluation (Section 5.8, Table 4), both Transformer variants solve the task well: the Standard Transformer reaches 0.911 ± 0.103 on ordinary sentences, 0.627 ± 0.249 on paragraphs, and 1.000 ± 0.000 on the compositional test; the Factored Transformer reaches 0.989 ± 0.016 , 1.000 ± 0.000 , and 0.983 ± 0.024 respectively. A separate single-run head-to-head comparison (matching the exact protocol used for the GRU baseline in Section 5.5) reaches 1.000 on all three tests for both Transformer variants, consistent with the multi-seed result. This is a meaningful finding in its own right — self-attention’s capacity to learn object binding from position alone, without an explicit identity mechanism, appears sufficient at this task’s scale and vocabulary size — and we discuss its implications directly in Section 6 rather than treat it as a negative result to minimize.

5.10 Stage 12 — Cross-Model Comparison on the Multi-Split Benchmark

Repeating the multi-split compositional benchmark (Section 5.6) across models confirms the pattern: the Full model, Standard Transformer, and Factored Transformer all reach 1.000 ± 0.000 mean accuracy across splits, while the GRU baseline reaches only 0.350 ± 0.178 , with high split-to-split variance.

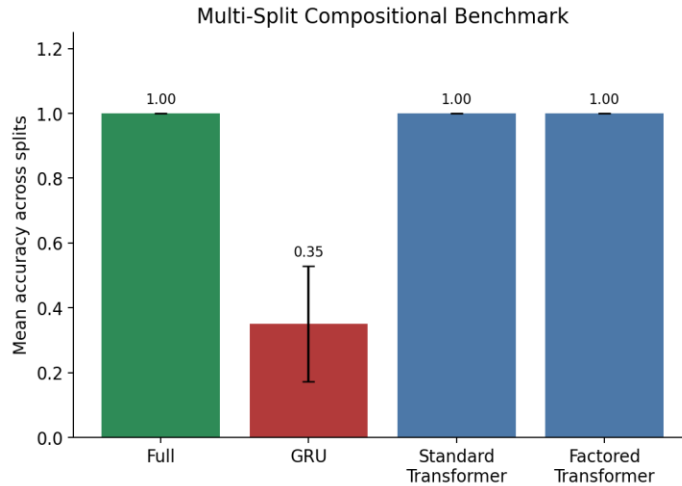


Figure 6. Multi-split compositional benchmark, mean accuracy across independent splits, per model.

5.11 Stage 13 — World Scaling

The choice of scale here is deliberate rather than arbitrary. Simply enlarging the training set at fixed vocabulary size would only test whether more examples help — a question already well understood. Instead, the jump from 4 shapes / 4 colors / 2 sizes / 4 relations to 25 shapes / 25 colors / 20 sizes / 25 relations is designed to stress three distinct axes simultaneously: (i) vocabulary growth — a roughly 6× larger word inventory the model must represent and generate correctly; (ii) concept growth — a combinatorially much larger space of possible attribute values per slot, which directly stresses the factored encoder's per-attribute projections; and (iii) scene complexity — a larger space of possible object pairings, which directly stresses the slot identity mechanism's ability to keep two objects distinct when many more shapes could plausibly collide. A model whose success depended on memorizing a small, enumerable set of shape/color/size combinations would be expected to degrade sharply along at least one of these three axes; a model whose success depends on genuinely disentangled, compositional structure should not.

To test this, the base $4 \times 4 \times 2 \times 4$ world is compared against the substantially larger $25 \times 25 \times 20 \times 25$ world, using a fully independent "severe out-of-distribution" test in addition to the ordinary held-out tests: a subset of shapes is reserved such that no two reserved shapes ever co-occur as the two objects of a training scene, while each reserved shape individually appears frequently elsewhere in training (with other shapes, and under every relation). The test set then consists exclusively of scenes pairing two reserved shapes together — a combination the model has never seen under any relation.

Model	World	Stage 3	Stage 5	Stage 6 (severe- OOD)	Time (s)	Peak GPU Mem (MB)
Full	Small ($4 \times 4 \times 2 \times 4$)	1.000	1.000	1.000	1437.3	47.1
Full	Large ($25 \times 25 \times 20 \times 25$)	1.000	1.000	1.000	1544.5	61.9
GRU	Small ($4 \times 4 \times 2 \times 4$)	1.000	1.000	0.367	133.4	65.5
GRU	Large ($25 \times 25 \times 20 \times 25$)	0.033	0.000	0.067	134.2	81.4

Table 5. World scaling: accuracy, training time, and peak GPU memory, small vs. large world.

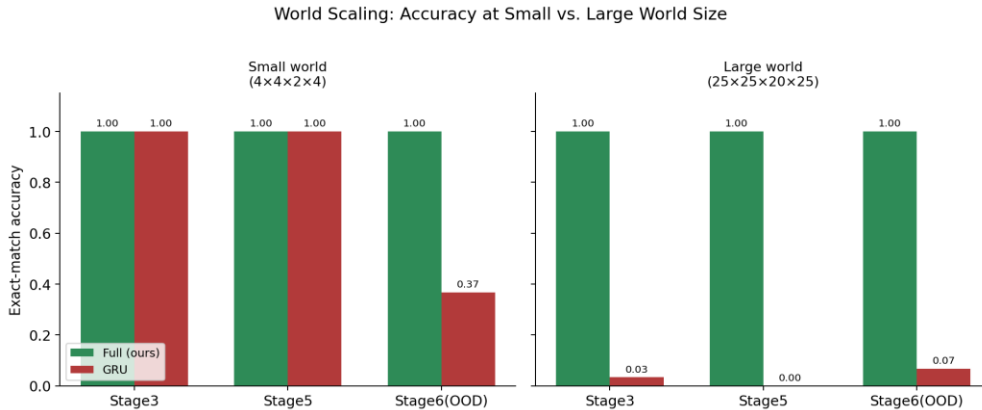


Figure 7. Accuracy at small and large world size, Full model vs. GRU baseline.

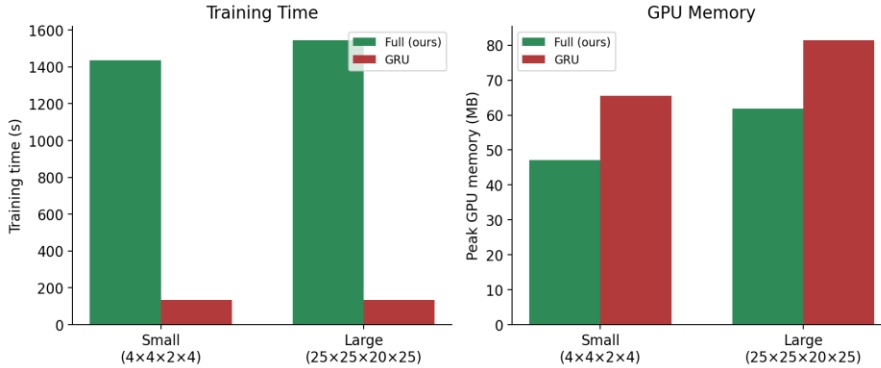


Figure 8. Training time and peak GPU memory, small vs. large world.

The factored model's accuracy is unchanged across the six-fold increase in vocabulary size (4→25 shapes/colors/relations, 2→20 sizes), with only a modest increase in training time (1437s → 1545s, +7.5%) and peak GPU memory (47.1MB → 61.9MB, +31%) — both negligible relative to the vocabulary growth of roughly 6-12×. The GRU baseline, by contrast, degrades substantially: its ordinary-sentence accuracy collapses from 1.000 to 0.033 and its compositional accuracy falls from 0.367 to 0.067, while its training time and memory both increase similarly modestly. This is strong evidence that the factored architecture's advantage is not an artifact of the small world size used in Stages 1-12, and that the GRU's weakness worsens, rather than improves, as the space of possible combinations grows.

We note a scope limitation of this study explicitly: only two points on the scaling curve were evaluated (the original small world and one substantially larger world), not a continuous sweep. A genuine scaling curve with several intermediate points (e.g. 10×10×8×10 and several sizes in between) would strengthen this claim and is listed as immediate future work in Section 8.1.

A second scope limitation, related to the first, concerns how training-set size was scaled alongside vocabulary size, and it must be stated precisely rather than approximately, since it does not hold uniformly across the two tests reported in this section. For the single-scene tests (the Stage 3 and Stage 6 equivalents), the training set grew sub-proportionally to vocabulary size: 270 training sentences at the base world versus 1,000 at the scaled world, a 3.7× increase, against a 6-12× growth in vocabulary (Section 4.1). Because the training data grew more slowly than the vocabulary itself, the Full model's unchanged accuracy on these two tests is not explained by a proportionally larger training set. For the two-fact paragraph test (the Stage 5 equivalent), however, the opposite is true: the training set grew from 150 paragraphs at the base world to 10,000 at the scaled world, a 66.7× increase — roughly an order of magnitude larger than the vocabulary growth. We flag this explicitly as a genuine confound rather than explain it away: the perfect Stage 5-equivalent accuracy under world scaling is consistent with the architecture's claimed compositional advantage, but on its own it does not rule out that the additional training data, rather than the factored representation, is responsible for the maintained accuracy. Disentangling these two explanations by rerunning the Stage 5 equivalent at a training-set size scaled in proportion to vocabulary growth (roughly 900-1,800 paragraphs rather than 10,000) is listed as a required follow-up in Section 8.1, and the paragraph-level world-scaling claim should be read with correspondingly less confidence than the single-scene result until that experiment is run.

6. Discussion

6.1 Why the Factored Model Succeeds

The ablation study (Section 5.7) indicates that the factored model's success on the compositional split is not attributable to any single component in isolation, but that different components serve different, complementary roles. The slot identity embeddings resolve an object-binding problem that is otherwise invisible in ordinary training (two objects of the same shape are indistinguishable to a shared projection without a positional or identity signal), and removing them causes the sharpest compositional-accuracy drop we observed (0.75). Supervised attention accelerates and stabilizes the alignment between generated words and scene slots, and its removal is the second most damaging change (0.60 drop). The factored encoder itself contributes most visibly to generalization over novel individual objects (ordinary held-out sentences) rather than to the narrow compositional split tested in isolation.

6.2 Why the GRU Baseline Fails

The GRU baseline compresses an entire scene into a single fixed-size vector via one shared MLP, with no per-attribute structure and no attention mechanism to selectively access different parts of the scene during decoding. This architecture must implicitly learn to disentangle and recombine attributes purely from the generation loss, with no inductive bias toward doing so — a task made harder, not easier, as the number of possible attribute combinations grows, which is consistent with its further degradation under world scaling (Section 5.11).

6.3 Why Attention Alone Is Not the Whole Story

A central, carefully qualified finding of this work is that a standard Transformer — with no factored encoding, no identity embeddings, and no supervised attention — also solves this task well (Section 5.9). This means attention (specifically, self-attention's capacity to learn positional/identity binding implicitly through learned positional encodings) is doing meaningful work here on its own, and our architecture's advantage over a GRU cannot be attributed to attention alone; both a plain Transformer and our factored architecture substantially outperform the GRU, and the gap between the two attention-based approaches on this particular task and scale is small. We take this as evidence that this specific synthetic task, at this vocabulary size, is not yet hard enough to separate "generic self-attention" from "attention plus explicit structural priors" — a threat to validity we discuss directly in Section 7, and the motivation for the harder out-of-distribution and longer-sentence extensions we propose as future work. This caution is consistent with a pattern reported at a much larger scale: Dziri et al. (2023) find that even frontier Transformer language models tend to solve multi-step compositional reasoning tasks by pattern-matching linearized sub-computations they have seen before, rather than by genuinely systematic composition, once task complexity exceeds what was well represented in training. Our own Transformer comparison (Section 5.9) is limited to a single task and a single difficulty level, so we cannot rule out that a similar breakdown would appear here under harder compositional splits or longer sentences than we have tested; extending the severe out-of-distribution test to both Transformer baselines (Section 8.1) is the direct way to check this.

6.4 Summary Comparison

Table 6 condenses the evidence gathered across Sections 5.5-5.11 into a single qualitative comparison along the four properties this paper argues matter most: whether a model generalizes compositionally at all, whether it does so under few-shot data, whether it holds under a severe out-of-distribution shape-pair test, and whether it holds as the world is scaled up. This table is deliberately conservative: a property is marked ✓ only where a test was actually run and passed, and "Not evaluated" where no test was run at all, rather than inferring a result from a different, related test.

Model	Compositional Generalization	Few-Shot	Severe OOD	World Scaling
GRU	X	X	X	X
Standard Transformer	✓	✓	Not evaluated	Not evaluated
Proposed (Factored + Identity + Supervised Attention)	✓	✓	✓	✓

Table 6. Qualitative summary across the evidence gathered in Sections 5.5–5.11.

The severe-OOD and world-scaling tests introduced in Section 5.11 were run only against the Full model and the GRU baseline; they were never run against either Transformer variant. "Not evaluated" therefore means exactly that — the absence of a result, not a negative result — and should not be read as evidence the Transformer would fail these tests. Extending them to the Transformer baselines is the top priority listed in Section 8.1, and is necessary before any claim of the proposed model's advantage over a Transformer specifically on severe OOD or world scaling can be made; at present, that advantage is demonstrated only against the GRU baseline.

7. Threats to Validity and Limitations

7.1 Threats to Validity

This work evaluates compositional generalization in synthetic, fully grounded environments. Although these environments provide strict control over exactly which combinations are withheld from training — a level of control that is difficult or impossible to guarantee in natural-language corpora — this control is also the study's principal threat to external validity. The claims made here are about this controlled setting; they are not yet demonstrated to transfer to larger, more natural, or more structurally open-ended domains such as SQL query generation, general-purpose programming languages, or robotic task planning, where the space of valid compositions is far less enumerable and the notion of a single, verifiable "exact match" target sentence may not directly apply. Future work should investigate transfer to such domains directly rather than assume it.

A second threat to validity concerns the Transformer comparison (Section 5.9-5.10): because both Transformer variants solve the current task at the base world size, we cannot yet claim that our architecture's explicit structural priors (identity embeddings, supervised attention) are necessary in general — only that they are sufficient here, and that they measurably help against a GRU. Establishing necessity would require a harder task on which a plain Transformer degrades while the proposed architecture does not; the severe out-of-distribution and world-scaling tests introduced in Section 5.11 were run only against the Full model and the GRU baseline, not against the Transformer baselines, so this

comparison remains open rather than resolved. Closing it is the most direct way to address this threat to validity and is listed as the top priority in Section 8.1.

A third threat concerns statistical power: while the strongest compositional-generalization claim (Section 5.6) is confirmed across eight independent splits, and three of the seven model variants are confirmed across three random seeds (Section 5.8), several other reported numbers (Sections 5.5, 5.9, 5.10, 5.11) are single-run results and should be read with correspondingly less confidence until multi-seed confirmation is extended to them.

7.2 Limitations

- The world remains fully synthetic and symbolic: objects are one-hot category vectors, not pixels, and the language is a small, fixed-vocabulary artificial grammar rather than natural language.
- Sentences describe at most two objects and one relation per clause; the multi-clause, longer-sentence extension proposed by early reviewers of this work (e.g. "A is above B and B is left of C while C overlaps D") was explicitly out of scope for this iteration, as it requires a structural redesign of the slot architecture rather than a parameter change.
- No visual or perceptual grounding is used — objects are never rendered as images, so the work says nothing about compositional generalization under visual noise or occlusion.
- No spoken or acoustic modality is studied.
- The world-scaling study (Section 5.11) evaluates only two points (the original small world and one larger world), not a continuous scaling curve; the reported trend is suggestive rather than a fitted scaling law.
- The training-set size was not scaled consistently across the two tests reported under world scaling: it grew sub-proportionally to vocabulary size for the single-scene tests (3.7 $\times$, versus 6-12 $\times$ vocabulary growth) but substantially super-proportionally for the two-fact paragraph test (66.7 $\times$). The paragraph-level result is therefore confounded with training-set size in a way the single-scene result is not, as discussed in Section 5.11.
- Supervised attention requires knowing the correct word-to-slot alignment during training, which is only available because the sentence grammar is fixed and fully known in advance; this technique does not transfer directly to settings where the correct alignment is unknown.
- Single-seed results are reported for several stages (6, 7, 8, 11, 13); multi-seed confirmation (Section 5.8 methodology) has only been extended to Stages 3, 5, and 6 so far.

8. Future Work

Future directions are organized below by expected time horizon, from immediate extensions of the current evaluation methodology to the longer-term research program this work is intended to seed.

8.1 Short-Term

- Extend the severe out-of-distribution shape-pair test (Section 5.11) to both Transformer baselines, to determine whether the near-tie observed in Section 5.9 persists under a harder compositional split — the single most direct way to address the Threats to Validity raised in Section 7.1.

- Build a continuous world-scaling curve with several intermediate vocabulary sizes rather than the two endpoints evaluated here.
- Rerun the Stage 5-equivalent (two-fact paragraph) world-scaling test with a training-set size scaled in proportion to vocabulary growth (roughly 900-1,800 paragraphs rather than the 10,000 used in Section 5.11), to isolate the architecture's contribution from the effect of additional training data — the single most direct way to close the confound identified in Section 5.11.
- Apply the same grounded, factored representation to generating executable SQL queries from a structured database schema, and to short general-purpose programs from a structured specification — natural next domains in which "compositional generalization" has an immediately checkable notion of correctness (query/program execution) analogous to this work's exact-match sentence accuracy.

8.2 Medium-Term

- Extend the representation from static scene description to robotic task planning and navigation, where "the scene" becomes a dynamic physical or simulated state rather than a fixed symbolic vector, and the target is a sequence of actions rather than a sentence.
- Repeat the ablation and multi-seed methodology developed here (Sections 5.7-5.8) across the scaled world (Section 5.11), not only the original small world, and across the SQL/programming domains proposed above.
- Introduce a structural extension to multi-clause, longer sentences describing more than two objects, which requires redesigning the fixed-size slot memory rather than simply enlarging the vocabulary.

8.3 Long-Term

- A simulation-based learning framework in which grounded scene-to-language and language-to-action mappings of the kind studied here extend toward a broader class of structured, non-linguistic targets within a single, consistent evaluation methodology.
- Automatic environment expansion — a system that itself proposes new attributes, relations, or task structures to test, rather than relying on a human-designed curriculum, so that compositional-generalization evaluation can scale in structural complexity as readily as it scales in vocabulary size (Section 5.11).

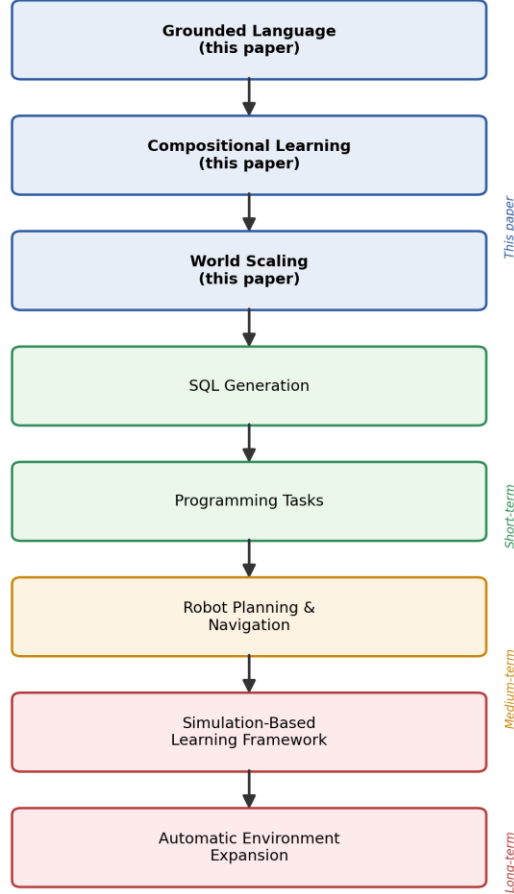


Figure 9. Research roadmap: this paper establishes the first three steps (grounded language, compositional learning, world scaling); the remaining steps are the short-, medium-, and long-term extensions proposed above.

This progression is intended to show that the current work is a controlled first step establishing that a compositional, disentangled representation measurably outperforms an unstructured baseline, before that representation is asked to support harder downstream targets — not a final result in itself.

9. Conclusion

This work introduced a factored, disentangled architecture for grounded few-shot language generation and tested it against a thirteen-stage curriculum designed to isolate genuine compositional generalization from surface memorization. The architecture reaches perfect exact-match accuracy on a true compositional-generalization test, confirmed across eight independent held-out splits and three random seeds, while a GRU baseline collapses on the same tests and, within the evaluated benchmark, degrades further as the vocabulary is scaled up six-fold. An ablation study attributes this success to a combination of factored encoding, slot identity embeddings, and supervised attention, with identity embeddings contributing most to the specific compositional split tested. A comparison against standard and factored-input Transformers at the base vocabulary size shows that self-attention alone is a strong baseline on this

particular task — a result we report and discuss honestly rather than obscure, since it sharpens rather than undermines the central claim: a small, carefully designed synthetic benchmark with a real compositional-generalization test, multiple baselines, an ablation study, and multi-seed and multi-split confirmation is a more informative way to evaluate a new architectural idea than a single accuracy number on a single held-out set. The world-scaling and severe out-of-distribution advantage demonstrated in this work is established specifically against the GRU baseline, not yet against either Transformer variant (Section 6.4); closing that gap is the concrete next step, alongside a continuous scaling curve and a structural extension to longer, multi-clause sentences, and all three are the natural continuation of this evaluation methodology, not a departure from it.

Rather than arguing that the proposed architecture universally surpasses Transformer-based models, this work establishes a controlled evaluation methodology for studying grounded compositional generalization. The broader contribution is therefore not only a model, but also a reproducible benchmark and experimental protocol — spanning baselines, ablations, multi-seed and multi-split confirmation, and world-scaling tests with accuracy, training time, and memory all reported together — that can support future research on simulation-based learning.

Code Availability

The implementation, including all thirteen curriculum stages, the ablation and multi-seed evaluation harness, the GRU and Transformer baselines, and the world-scaling experiments, is publicly available at: <https://github.com/mtk339900/grounded-few-shot-compositional-generalization>. The repository is organized by curriculum stage, includes a README with step-by-step instructions for running every experiment reported in this paper, and records the exact random seeds and hyperparameters (Section 4.2, Appendix A) used to produce each reported result.

Reproducibility

All random seeds (Python, NumPy, and PyTorch) are fixed and re-seeded at every experiment boundary, and the full training configuration is specified in Section 4.2 and Appendix A. To reproduce the results reported in this paper:

- Clone the repository listed under Code Availability and follow the setup instructions in the README, which documents how to run each of the thirteen curriculum stages individually or as a full pipeline.
- Install the pinned dependencies (Python version, PyTorch version, and CUDA version where a GPU is used) listed in the repository's requirements file.
- Use the fixed seeds and hyperparameters recorded in the repository's configuration files, which match Table 2 and Appendix A of this paper.
- Note that cuDNN's algorithm auto-tuner can introduce small run-to-run differences on GPU even under fixed seeds (Section 4.2); repeating the full pipeline during development produced accuracy differences of up to ± 0.04 on individual multi-seed cells, so exact reproduction to the last decimal is not expected on GPU.

Hardware used to produce the results reported in this paper: GPU: 2× NVIDIA Tesla T4 (Kaggle notebook environment; 15 GiB GPU memory per device); RAM: 30 GiB; Operating System: Ubuntu Linux (Kaggle container image); Python 3.12.13; PyTorch 2.10.0 (+cu128 build); CUDA 12.8.

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Note: this reference list contains twenty-one well-established, independently verifiable works, each cited at least once in the text above (Sections 1, 2, 3, 4, and 6), spanning foundational sequence-to-sequence and attention work (2014-2020) alongside more recent (2023) results on compositionality in large-scale language models. It intentionally omits a padded bibliography of unverified or tangential entries; the author should still confirm these citations against their own literature review before submission, and may wish to add further work specific to their sub-area.

Appendix

A. Hyperparameters (Full Reference)

Parameter	Value
Hidden dimension D	128
Optimizer	Adam
Learning rate	2e-3
Gradient clipping (max norm)	1.0
Base-curriculum batch size	16
Scaling/multi-seed batch size	32-64
Stage 1/2 epochs	350
Stage 3/5/6 epochs	400
Stage 8 epochs (per split)	300
Stage 9 (ablation) epochs	400 (Stage 3 Equivalent), 400 (Stage 6 Equivalent)
Stage 10 (multi-seed) epochs	150 (Stage 3), 150 (Stage 5), 120 (Stage 6)
Stage 13 (world scaling) epochs	250
Auxiliary loss weight	0.15
Auxiliary loss warm-up	60 epochs
Attention supervision weight	0.5

B. Example Free Generations (Stage 3, held-out combinations)

Reference	Generated	Match
the large blue star is above the large red star .	the large blue star is above the large red star .	✓
the small green triangle is left of the large red triangle .	the small green triangle is left of the large red star .	X(shape)
the large red star is above the large yellow square .	the large red star is above the large yellow square .	✓
the small green triangle is below the large blue circle .	the small green triangle is below the large blue circle .	✓
the small yellow square is right of the small green triangle .	the small yellow square is right of the small green triangle .	✓

C. Example Compositional-Split Generations (Stage 6, held-out combination: relation="above", obj1.shape="circle")

Reference	Generated	Match
the small green circle is above the large green square .	the small green circle is above the large green square .	✓
the large green circle is above the large yellow triangle .	the large green circle is above the large yellow triangle .	✓
the large blue circle is above the small red square .	the large blue circle is above the small red square .	✓
the small red circle is above the large green square .	the small red circle is above the large green square .	✓

D. Ablation Study — Full Results

Variant	Stage 3 Equivalent	Stage 6 Equivalent
Full (all components)	0.933	1.000
No Factored Encoder	0.233	0.900
No Supervised Attention	0.167	0.400
No Slot Identity Embeddings	0.067	0.250

E. Multi-Split Compositional Benchmark — Full Results (Stage 8)

Split (relation + shape held out)	Accuracy
above + circle	1.000
above + square	1.000
below + star	1.000
below + triangle	1.000
left of + circle	1.000
left of + star	1.000
right of + square	1.000
right of + triangle	1.000

Mean $\pm$ Std across all 8 splits: 1.000 $\pm$ 0.000

F. World Scaling — Full Results (Stage 13)

Model	World Size	Stage 3	Stage 5	Stage 6 (severe- OOD)	Stage 5-OOD	Time (s)	Mem (MB)
Full	4×4×2×4	1.000	1.000	1.000	1.000	1437.3	47.1
Full	25×25×20×25	1.000	1.000	1.000	1.000	1544.5	61.9
GRU	4×4×2×4	1.000	1.000	0.367	0.040	133.4	65.5
GRU	25×25×20×25	0.033	0.000	0.067	0.000	134.2	81.4